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“Pink Tax: A Data-Driven Analysis of Gender-Based Price Differences in Indian Online Retail”

A Report

Submitted By

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Under the guidance of

Prof. Himani Patel

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2026



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This is to certify that the “**Pink Tax: A Data-Driven Analysis of Gender-Based Price Differences in Indian Online Retail**” has been carried out by **Kanishka Shah (231260131051)** under my guidance in completion of Summer Internship in Computer Science & Engineering Branch, 7th Semester of Gujarat Technological University, Ahmedabad during the academic Year 2026-27.

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ABSTRACT

The “Pink Tax” is more than a pricing issue. It raises a simple question: do products marketed to women actually cost more than comparable products marketed to men? This project uses data analysis to investigate gender-based price differences across everyday consumer products sold online in India. It compares similar items across categories, brands and product types, and all prices are in Indian Rupees (₹).

Three public Kaggle datasets were combined: product listings from **Myntra** (fashion, footwear and accessories), **BigBasket** (beauty and hygiene) and **Amazon.in** (personal care). After cleaning, de-duplication, outlier removal and tagging each product by its target gender using product URLs and English and Hindi keywords, the final dataset contained **79,390 products** from **2,769 brands** in **44 comparable categories**. The analysis was done in Python (Pandas, NumPy, SciPy, Statsmodels, scikit-learn, Matplotlib, Seaborn and Plotly) on Google Colab, with an Excel workbook and a Power BI dashboard for reporting.

The prices were examined through four lenses of increasing strictness. The first compared women's and men's prices within each category using non-parametric tests with bootstrap confidence intervals and false-discovery-rate correction. The second compared price per 100 ml for liquids. The third compared 860 brand-matched pairs, where the same brand sells a men's and a women's version in the same category. The fourth was a fixed-effects regression that controls for brand and category.

The results show that **there is no blanket Pink Tax** in Indian online retail. Women paid significantly more in only 4 of 44 categories, while men paid more in 22. Within the same brand, the women's version was dearer in 27% of pairs and the men's version in 54%, and the typical gap was -5.9%. Personal-care liquids showed no significant difference per 100 ml. The Pink Tax does, however, appear consistently in specific pockets: girls' night suits (+78%), innerwear (+33%), kids' clothing sets (+11%) and women's sports shoes (+5%). Women's deodorants on BigBasket also cost 17% more at the selling price because they are discounted less. Much of the gap comes from brand mix and product differentiation rather than a uniform surcharge. The practical lesson for consumers is to compare unit prices and to look across the gender aisle.

Keywords: Pink Tax, gender-based pricing, price discrimination, e-commerce, hypothesis testing, fixed-effects regression, data analytics, Python, Power BI.

Introduction

The term **Pink Tax** describes the extra amount women are said to pay for products and services that are marketed to them, when an almost identical version marketed to men costs less. A razor in pink packaging, a “for her” deodorant or a girl's T-shirt may cost more than its male counterpart even when the materials and function are the same. Unlike a real tax, the money does not go to the government. It is a price difference created by marketing and pricing decisions.

The issue became widely known after a 1994 study by the California Assembly, which led to the Gender Tax Repeal Act of 1995 for services. The best-known study of goods is the New York City Department of Consumer Affairs report *From Cradle to Cane* (2015). It compared nearly 800 products with clear male and female versions and found that women's products cost about 7% more on average, and 13% more for personal-care products. Later work has been more cautious. The U.S. Government Accountability Office (2018) found higher prices for women in 5 of 10 personal-care categories but could not attribute them to gender bias alone. Moshary, Tuchman and Bhatia (2021) found no systematic premium when truly identical products were compared.

In India, public debate has focused mainly on the “tampon tax”, which ended when sanitary napkins were exempted from GST in July 2018. Gender-based pricing of everyday goods has received far less data-driven attention, even though online retail now makes it possible to observe the prices of tens of thousands of products at once.

This project fills that gap with a reproducible data-analysis pipeline. It collects real product listings from three Indian e-commerce platforms, labels each product as targeted at women or men, and uses statistical tests, matched comparisons and regression to measure where the Pink Tax exists, how large it is and what explains it. The work was carried out during a 15-day summer internship. Chapter 1 introduces the internship, its aims and the tools used. Chapter 2 records the day-by-day activities and learning, and Chapter 3 presents the conclusions.

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Chapter 1: Introduction

1.1 Introduction of Summer Internship

The summer internship was a 15-day, project-based programme in **data analytics**. Its purpose was to take a real-world question from idea to evidence: finding and understanding data, cleaning and transforming it, analysing it with statistics, and communicating the results through charts, spreadsheets and dashboards. The work was organised as a single end-to-end project, *the Pink Tax in Indian online retail*, so that every new skill was applied straight away to the same dataset.

The internship was divided into four phases:

- **Foundations (Days 1–3):** problem understanding, literature review, setting up Python, Google Colab and GitHub, and revising NumPy and Pandas.
- **Data engineering (Days 4–7):** sourcing data from Kaggle, profiling it, cleaning it and engineering features such as gender, category and price per 100 ml.
- **Analysis (Days 8–13):** exploratory analysis, visualisation, hypothesis testing, brand-matched comparison, regression and a machine-learning check.
- **Reporting (Days 14–15):** an Excel workbook with live formulas, a Power BI dashboard, the GitHub repository and this report.

Each day ended with a short log of what was done, what was learned and what problems came up. Chapter 2 is based on these logs.

1.2 Aim & Objective

Aim: to find out, using real data and sound statistical methods, whether products marketed to women in India cost more than comparable products marketed to men, and where any such Pink Tax appears most consistently.

Objectives

1. Build a clean, gender-tagged dataset of Indian consumer products from public sources.
2. Compare women's and men's prices within each product category and test whether the differences are statistically significant.
3. Compare like with like: price per 100 ml for liquids, and the same brand's men's and women's lines.
4. Separate a true gender premium from brand and category effects using regression.
5. Check whether discounts change the gap between the list price (MRP) and the price actually paid.

6. Present the results in a Colab notebook, an Excel workbook and a Power BI dashboard, and publish the project on GitHub.

Research questions and hypotheses

- **RQ1:** Within a product category, do women's products have higher prices than men's?
 - **RQ2:** Does the gap survive when unit price, brand and category are held constant?
 - **RQ3:** In which categories does the Pink Tax appear most consistently?
 - **H₀:** the price distributions of women's and men's products are the same. **H₁:** they differ.
- The significance level is $\alpha = 0.05$, with a false-discovery-rate correction.

1.3 Tools & Technology used during Internship

Table 1.1: Tools and technologies used

Tool / Library	Purpose in this project
Python 3.11 with Google Colab / Jupyter	Main language and notebooks for the complete, reproducible data pipeline
NumPy and Pandas	Numerical operations, bootstrap resampling; loading, cleaning, grouping and pivoting data
Regular expressions (re)	Extracting gender, category and pack size from product titles and URLs
Matplotlib, Seaborn and Plotly	Publication-quality static charts and an interactive chart with hover details
SciPy and Statsmodels	Mann-Whitney, Welch, Wilcoxon and chi-square tests; OLS / fixed-effects regression, FDR correction
scikit-learn	Random-forest price model, target encoding, permutation importance
Microsoft Excel and OpenPyXL	Workbook with COUNTIFS / AVERAGEIFS summaries, pivot-style tables, conditional formatting
Power BI	Star-schema data model, DAX measures and an interactive dashboard
Git, GitHub and Kaggle	Version control, publishing (github.com/mppyxx/PinkTax) and the data source

1.4 Dataset Description

All data came from public Kaggle datasets, and every price is in Indian Rupees. The **MRP** (maximum retail price) is the list price printed by the brand. The **selling price** is the price after the platform's discount.

Table 1.2: Datasets used

Source	Raw size	Used	Content
Myntra (2023)	168,029 × 13	72,771	Apparel, footwear, accessories, perfume; men, women, boys, girls
BigBasket (2022)	27,555 × 10	736	Beauty & hygiene: deodorants and perfumes
Amazon.in (2023)	1,589,160 × 11	5,883	Personal care: deodorant, perfume, razor, trimmer, washes, shampoo
Total		79,390	44 categories, 2,769 brands; adults: 46,022 men's and 28,906 women's items; kids: 4,462 items

Chapter 2: What I Learned during Internship

Day-1: Orientation and Understanding the Pink Tax

Objective: understand the problem, review earlier research and turn a broad idea into questions that can be tested.

Activities performed

- Attended the orientation session and discussed the project scope: an end-to-end data analytics study of gender-based pricing.
- Read the NYC Department of Consumer Affairs study *From Cradle to Cane* (2015). Women's products cost 7% more on average, and 13% more for personal care.
- Read the U.S. GAO report (2018) and the paper by Moshary et al. (2021), which argue that many gaps disappear when truly identical products are compared.
- Wrote down the research questions (RQ1–RQ3) and the null and alternative hypotheses.
- Planned the workflow shown in Figure 2.1.

Learning outcome

Good analysis starts with a precise question. “Do women pay more?” is too vague. “Within the same category and brand, is the women's version priced higher?” can be tested. I also learned the difference between the **Pink Tax** (a price gap between gendered versions of a product) and the **tampon tax** (a sales tax on menstrual products, removed in India in 2018).

Project workflow: from raw listings to evidence



Figure 2.1: Project workflow planned on Day 1

Day-2: Setting up the Development Environment

Objective: prepare a reproducible working environment for the project.

Activities performed

- Set up Google Colab and a local Jupyter environment, and learned the difference between a hosted runtime and a local kernel.
- Installed and imported the main libraries: NumPy, Pandas, Matplotlib, Seaborn, Plotly, SciPy, Statsmodels, scikit-learn and OpenPyXL. Recorded their versions in *requirements.txt*.
- Created the GitHub repository **PinkTax** and a clear folder structure: *src/*, *data/*, *notebooks/*, *outputs/*, *excel/*, *powerbi/* and *report/*.
- Learned basic Git commands (*clone*, *add*, *commit*, *push*) and wrote a *.gitignore* so that large raw files are not uploaded.
- Wrote *config.py* to keep all paths and chart colours in one place.

Learning outcome

A project is only useful if someone else can run it. Keeping the code in numbered scripts (*01_download_data.py ... 07_powerbi_export.py*) and the settings in one configuration file means the whole analysis can be reproduced with a few commands, locally or in Colab.

```
In [2]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
import statsmodels.formula.api as smf
from IPython.display import Image, display

from config import RAW, PROCESSED, TABLES, FIGURES, GENDER_COLORS, set_style
set_style()
pd.set_option("display.max_columns", 30)
pd.set_option("display.width", 160)
print("pandas", pd.__version__, "| numpy", np.__version__, "| seaborn", sns.__version__)

pandas 3.0.6 | numpy 2.4.6 | seaborn 0.13.2
```

Figure 2.2: Importing the analysis libraries in the Colab notebook

The Pink Tax in Indian Online Retail — Data Analysis Notebook

[Open In Colab](#)

Question: do products marketed to women cost more than comparable products marketed to men?

Data: three public Kaggle datasets with prices in Indian Rupees (₹): Myntra (fashion), BigBasket (beauty & hygiene) and Amazon.in (personal care).

Libraries used: NumPy · Pandas · Matplotlib · Seaborn · Plotly · SciPy · Statsmodels · Scikit-learn · OpenPyXL

Step	What happens
1	Setup and data download
2	First look at the raw data
3	Cleaning, gender tagging and category tagging
4	Exploratory data analysis (EDA)
5	Hypothesis testing
6	Brand-matched comparison
7	Regression with fixed effects
8	Machine-learning check
9	Interactive chart, Excel and Power BI export

Figure 2.3: Notebook overview with the Open-in-Colab link

Day-3: Python Refresher – NumPy and Pandas

Objective: strengthen the Python data-handling skills needed for the project.

Activities performed

- Revised NumPy arrays, vectorised operations, broadcasting and random number generation, which are used later for bootstrapping.
- Practised Pandas: Series and DataFrame, *loc/iloc* indexing, boolean filtering, *groupby*, *agg*, *pivot_table*, *merge* and *concat*.
- Learned the *.str* accessor for text columns: *contains*, *extract*, *replace* and *split*.
- Practised regular expressions such as *-(men|women|boys|girls)-*, which later pulled the gender out of Myntra product URLs.
- Solved small exercises: average price per brand, the most expensive category, and the share of discounted items.

Learning outcome

Vectorised Pandas code is much faster and clearer than Python loops. Filtering about 1.6 million Amazon rows took seconds with boolean masks. Regular expressions are a powerful tool for turning free text into structured features.

Table 2.1: Pandas operations practised and where they were used

Operation	Used for
<code>read_csv</code> , <code>info</code> , <code>describe</code>	Loading and profiling the raw files
<code>str.extract</code> / <code>str.contains</code>	Gender and category tagging, pack-size extraction
<code>drop_duplicates</code> , <code>dropna</code>	Cleaning
<code>groupby</code> + <code>agg</code> / <code>transform</code>	Category medians, outlier limits, fixed effects
<code>pivot_table</code> , <code>crosstab</code>	Summary tables and brand-matched pairs

Day-4: Data Sourcing from Kaggle

Objective: find public datasets with real Indian prices that can be tagged by gender.

Activities performed

- Searched Kaggle for “pink tax”, “gender price”, Myntra, Nykaa, BigBasket, Flipkart and Amazon India. No ready-made Pink Tax dataset exists, so product catalogues had to be used.
- Evaluated candidates against four criteria: prices in ₹, a way to identify the target gender, enough products per category, and a clear licence and source.
- Selected three datasets: **Myntra 168k products**, the **BigBasket product list** and **Amazon India products 2023** (1.5 million listings).
- Wrote `01_download_data.py` to download and unzip the files automatically through the Kaggle API, so that the download is reproducible.
- Loaded the files and checked their shapes and columns (Figure 2.4).

Learning outcome

In real projects the perfect dataset rarely exists. The skill is to combine sources that each answer part of the question. Myntra covers fashion with gender in the URL, BigBasket labels men's and women's deodorants, and Amazon titles include pack sizes for unit-price comparison.

```
In [4]: myntra = pd.read_csv(RAW / "myntra_products.csv")
bigbasket = pd.read_csv(RAW / "bigbasket_products.csv")
print("Myntra : ", myntra.shape)
print("BigBasket:", bigbasket.shape)
myntra.head(3)
```

Myntra : (168029, 13)
BigBasket: (27555, 10)

```
Out[4]:
```

	product_name	brand_name	rating	rating_count	marked_price	discounted_price	sizes	product_link
0	Croc Textured Two Fold Wallet	Lino Perros	0.0	0	1295	828	Onesize	wallets/liino-perros/liino-perros-women-peach-co... https://assets
1	Men Striped Sliders	Mast & Harbour	4.0	76	1299	584	UK6,UK7,UK8,UK9,UK10,UK11	flip-flops/mast-harbour/mast-harbour-men-nav... https://assets
2	Printed A-line Kurta	Biba	4.3	66	1999	1599	S,M,L,XL,XXL,3XL	kurtas/biba/biba-women-off-white-black-printe... https://assets

Figure 2.4: Loading the raw Myntra and BigBasket files with Pandas

Day-5: Data Understanding and Profiling

Objective: understand what each column means, how the data is stored and what quality problems exist.

Activities performed

- Used `info()`, `describe()` and `value_counts()` to check data types, missing values and distributions.
- Found where the gender is stored in each source: in the product URL for Myntra, in the product type for BigBasket and only in the title for Amazon.
- Found that most Amazon titles are in **Hindi** (for example “पुरुषों के लिए” means “for men”), so English-only keywords would miss most products.
- Noted quality issues: zero list prices, duplicate listings, unisex and combo packs, and extreme prices.
- Decided to use MRP as the main price and the selling price as a second measure.

Learning outcome

Profiling before cleaning saves time. Checking gender labels on Myntra showed 58,436 women's, 53,615 men's, 4,671 girls' and 4,209 boys' products, plus 43,823 without a gender tag. That told me how much usable data there was before writing any cleaning code.

```
In [6]: # The gender is hidden inside Myntra's product URL, e.g. ".../puma-men-black-sneakers/..."
myntra["gender_raw"] = myntra["product_link"].str.extract(r"-(men|women|boys|girls|unisex)-")[0]
myntra["gender_raw"].value_counts(dropna=False)

Out[6]: gender_raw
women    58436
men      53615
NaN      43823
girls     4671
boys      4209
unisex    3275
Name: count, dtype: int64
```

Figure 2.5: Extracting the target gender from Myntra product URLs using a regular expression

Day-6: Data Cleaning

Objective: turn the raw files into a reliable, analysis-ready table.

Activities performed

- Removed duplicate listings using product links, ASINs, or name + brand + price.
- Dropped products with zero or missing prices, and capped the selling price at the MRP.
- Removed unisex, “men and women”, couple and multipack products, because they have no single target gender.
- Kept only categories with at least 15 products for **each** gender, so that every comparison is between genuine alternatives.
- Removed outliers beyond $3 \times \text{IQR on the log price}$ within each category, a rule that suits right-skewed prices.
- Recorded every step in a cleaning log (Table 2.2).

Table 2.2: Cleaning log

Step	Rows remaining
Products in scope (Myntra 72,914 + BigBasket 7,273 + Amazon 62,078)	142,265
After gender and category tagging	80,008
After removing zero or missing prices	79,817
After keeping comparable categories (≥ 15 per gender)	79,553
After log-IQR outlier removal and final de-duplication	79,390

Learning outcome

Every cleaning rule is a decision that can change the result, so each rule must be justified and documented. Using the log scale for outliers was important. A normal IQR rule on raw prices would have removed many genuine premium products.

```
In [10]: pd.read_csv(PROCESSED / 'cleaning_log.csv')
Out[10]:
```

	source	step	rows
0	Myntra	raw rows in scope	72914
1	BigBasket	raw rows in scope	7273
2	Amazon.in	raw rows in scope	62078
3	all	after gender + category tagging	80008
4	all	after removing zero / missing prices	79817
5	all	after keeping comparable categories (>=15 per ...	79553
6	all	after log-IQR outlier removal	79391

Figure 2.6: Cleaning log produced by the pipeline

Day-7: Feature Engineering – Gender, Category and Unit Price

Objective: create the variables needed for a fair comparison.

Activities performed

- **Gender:** built keyword rules in English (men, man's, him, gents, women, her, ladies, femme ...) and Hindi (पुरुष, महिला). Titles that match both genders are excluded.
- **Category:** wrote ordered regular-expression rules. For example, *attar* is tested before *razor* because the Hindi word for “treasure” (देज़र) contains the letters of “razor” (रेज़र). I found and fixed this bug during spot checks.
- **Pack size and unit price:** extracted ml and g values from titles and calculated ₹ per 100 ml for liquids.
- **Discount %** = $(1 - \text{selling price} / \text{MRP}) \times 100$.
- **Segment:** grouped categories into Apparel, Footwear, Accessories and Personal Care, and added an age group (Adult or Kids).
- Saved the tidy master table (79,390 rows $\times$ 16 columns) as *pinktax_master.csv*.

Learning outcome

Automatic tagging must always be checked by reading real examples. Printing random samples for each category and gender showed mistakes that summary statistics would never reveal, such as hair-removal creams being counted as razors.

```
In [8]: %run ../src/02_clean_and_tag.py
```

source	step	rows
Myntra	raw rows in scope	72914
BigBasket	raw rows in scope	7273
Amazon.in	raw rows in scope	62078
all	after gender + category tagging	90808
all	after removing zero / missing prices	79817
all	after keeping comparable categories (>=15 per gender)	79553
all	after log-IQR outlier removal	79391

Master table: 79,390 products			
gender	Men	Women	
source			
Amazon.in	3785	2098	
BigBasket	469	267	
Myntra	44451	28320	

source	age_group	category	Men	Women
Amazon.in	Adult	Body Wash	97	28
		Deodorant	766	377
		Face Wash	95	31
		Hair Removal Cream	16	25
		Perfume	2238	1378
		Razor	157	83
		Shampoo	60	25
		Trimmer	356	151
BigBasket	Adult	Deodorant	329	171
		Perfume	140	96
Myntra	Adult	Belts	564	250
		Blazers	356	143
		Casual Shoes	3287	1066
		Flip-Flops	1014	845
		Innerwear (Briefs)	408	488
		Jackets	1055	814
		Jeans	3946	2621
		Kurtas	792	6023
		Lounge Pants	189	399
		Night Suits	90	1351
		Perfume	327	178
		Shirts	9282	922
		Shorts	1533	805
		Socks	268	84
		Sports Shoes	1988	882
		Sunglasses	100	119
		Sweaters	174	283
		Sweatshirts	673	370
		T-Shirts	9383	3778
		Track Pants	1885	918
		Tracksuits	78	74

Figure 2.7: Output of the cleaning and tagging script: products per source, category and gender

Day-8: Exploratory Data Analysis (EDA)

Objective: understand the shape of the data before testing anything.

Activities performed

- Built pivot tables of product counts and median prices by source, segment and gender.
- Checked the distributions. Prices are strongly right-skewed, so medians and log prices are better summaries than means.
- Compared means with medians. In personal care the mean women's price is 21% higher, but the median is 7% lower, because a few expensive women's perfumes pull the mean up.
- Found that the sample is unbalanced: Myntra lists far more men's shirts and T-shirts, and far more women's kurtas and night suits.

Learning outcome

EDA stops the analysis from reaching the wrong conclusions. Comparing overall averages would be misleading because men's and women's catalogues contain different mixes of products. Every later comparison was therefore made **within** a category.

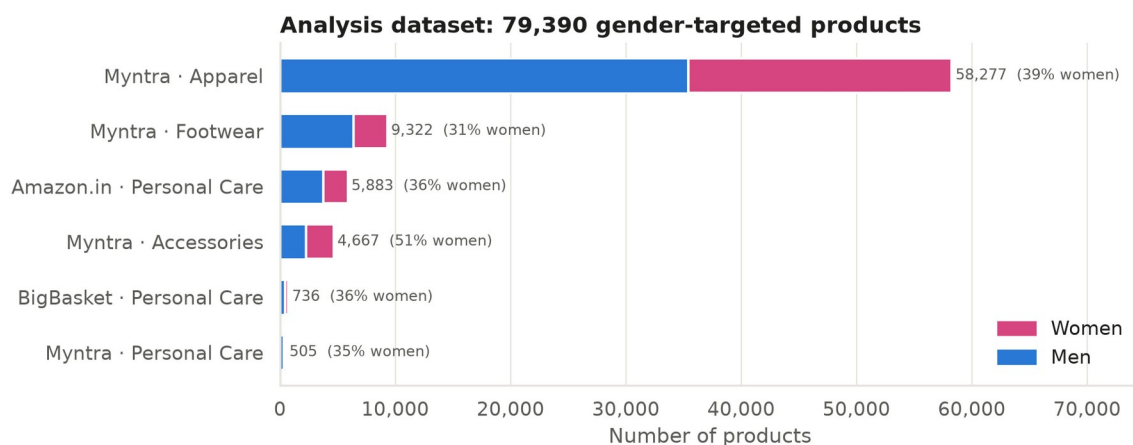


Figure 2.8: Composition of the final dataset by source, segment and gender

```
In [11]: df.pivot_table(index=["source", "segment"], columns="gender", values="mrp",
aggfunc=["count", "median"]).round(0)
```

Out[11]:

		count		median	
		Men	Women	Men	Women
source	segment				
Amazon.in	Personal Care	3785	2098	774.0	699.0
BigBasket	Personal Care	469	267	348.0	500.0
Myntra	Accessories	2290	2377	2699.0	2799.0
	Apparel	35424	22853	1899.0	1790.0
	Footwear	6410	2912	2597.0	1899.0
	Personal Care	327	178	1499.0	1798.0

Figure 2.9: Pivot table of product counts and median MRP

Day-9: Data Visualisation with Matplotlib, Seaborn and Plotly

Objective: design clear and honest charts that explain the data.

Activities performed

- Chose one colour for women (pink) and one for men (blue) and checked that the pair can be told apart by colour-blind readers. The same colours are used in Python, Excel and Power BI.
- Drew box plots on a log scale (Figure 2.10), grouped bar charts, dot plots with confidence intervals, histograms and a correlation heatmap.
- Followed chart design rules: direct labels, a single axis, light grid lines, and titles that state what the chart shows.
- Built an interactive Plotly chart where hovering over a category shows the sample sizes, medians and p-value.

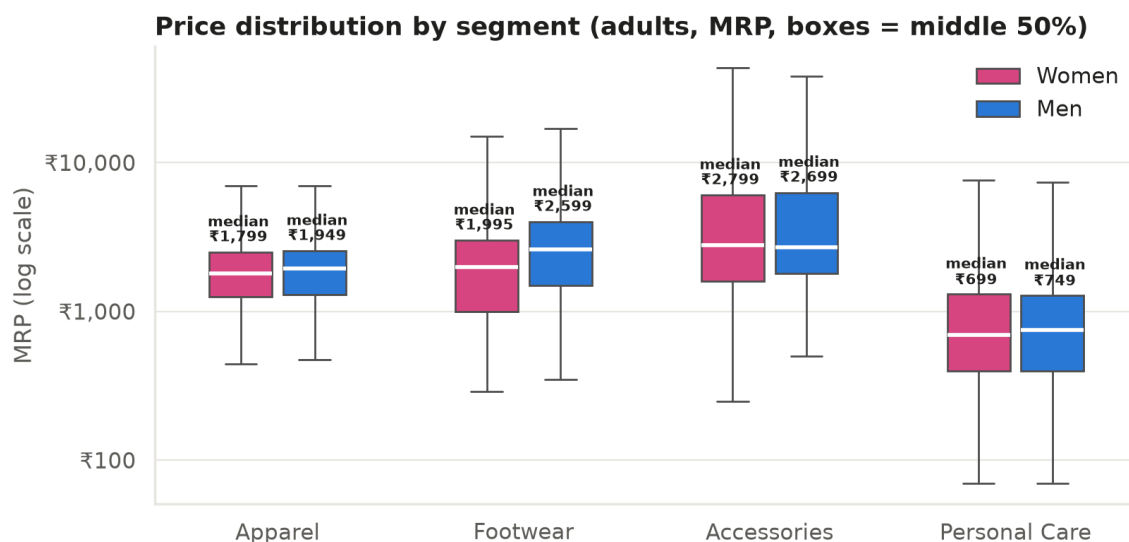


Figure 2.10: MRP distribution by segment and gender (adults)

Learning outcome

A chart is an argument. The log scale made both cheap socks and expensive watches readable on one axis. The box plots already hint that adult women's median prices are not higher than men's in apparel and footwear.

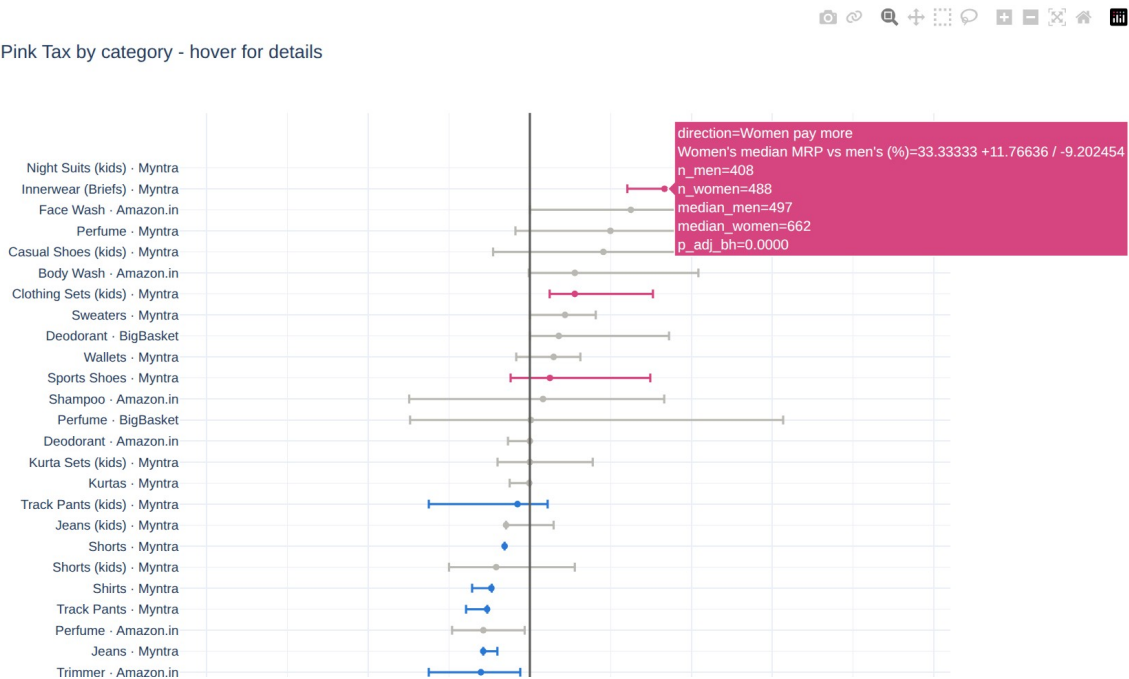


Figure 2.11: Interactive Plotly chart with hover details

Day-10: Hypothesis Testing Across 44 Categories

Objective: test, category by category, whether women's and men's prices differ.

Activities performed

- Used the **Mann-Whitney U test**, which does not assume a normal distribution, and **Welch's t-test on log price**, which allows unequal variances.
- Measured the effect size with **Cohen's d** and computed **bootstrap 95% confidence intervals** (2,000 resamples) for the median premium.
- Corrected for testing 44 categories at once with the **Benjamini-Hochberg** false-discovery-rate method.
- Worked example: adult T-shirts on Myntra had a median of ₹999 for women vs ₹1,299 for men (-23% , $p < 10^{-100}$).

Result

Women pay significantly more in **4 of 44** categories: girls' night suits (+78%), innerwear (+33%), kids' clothing sets (+11%) and sports shoes (+5%). Men pay more in **22** categories, including belts (−43%), blazers (−36%), sunglasses (−32%) and jackets (−25%). The other 18 categories show no significant gap.

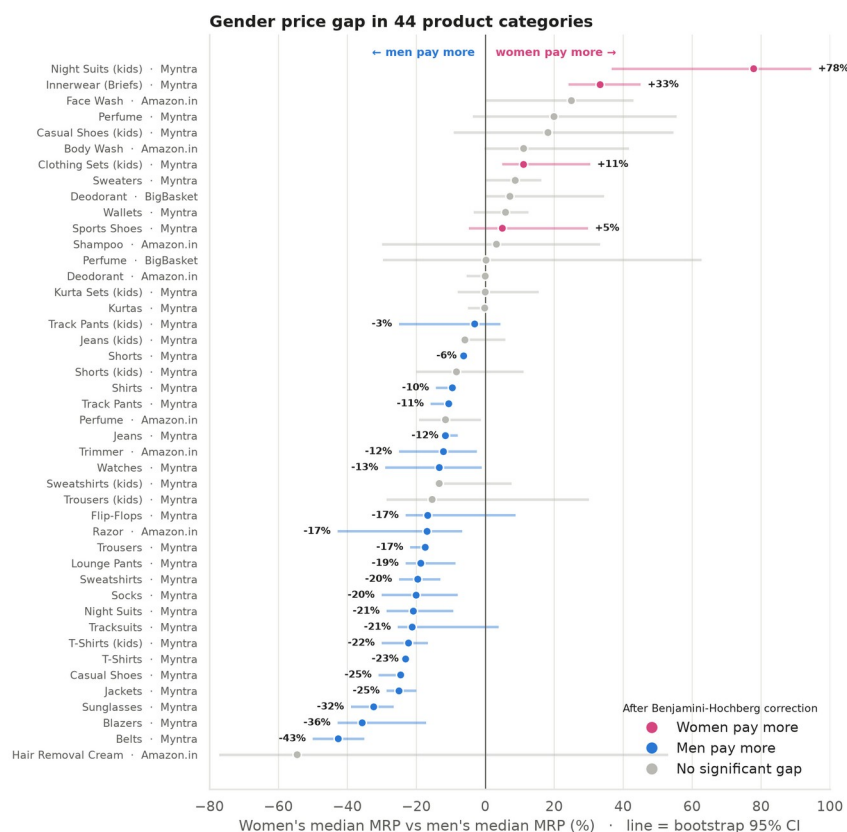


Figure 2.12: Median price gap for 44 categories with 95% confidence intervals

Day-11: Comparing Like with Like – Unit Price and Brand-Matched Pairs

Objective: remove the effect of pack size and brand from the comparison.

Activities performed

- Compared ₹ per 100 ml for deodorants, perfumes, body wash, face wash and shampoo. No category showed a significant gap. Deodorant, for example, was ₹210 vs ₹201 per 100 ml (+4.6%, $p = 0.80$).
- Built **860 brand-matched pairs**, each one brand selling both a men's and a women's line in the same category, and compared the two median prices.
- Tested the pairs with the **Wilcoxon signed-rank test** and a **sign test**, and used a chi-square test to check whether the outcomes differ by segment.

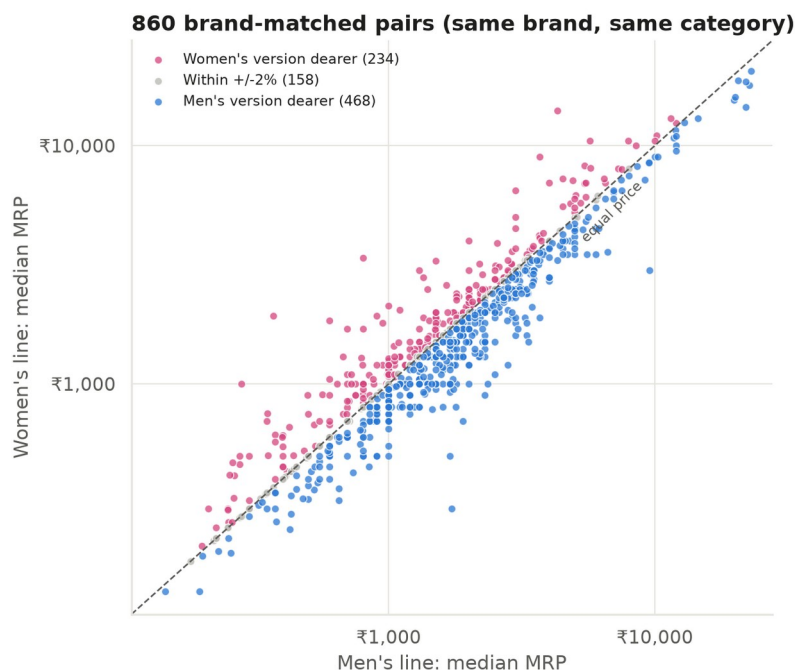


Figure 2.13: Brand-matched pairs. Points above the line mean the women's line is dearer

Result

The women's version was dearer in **27%** of pairs, the men's in **54%**, and 18% were within $\pm 2\%$. The typical gap was **-5.9%** (95% CI -7.7% to -4.0%, $p < 0.001$). Personal care was the only balanced segment (35% vs 37%, $p = 0.93$), and the outcome depends on the segment ($\chi^2 = 37.0$, $p < 0.001$).

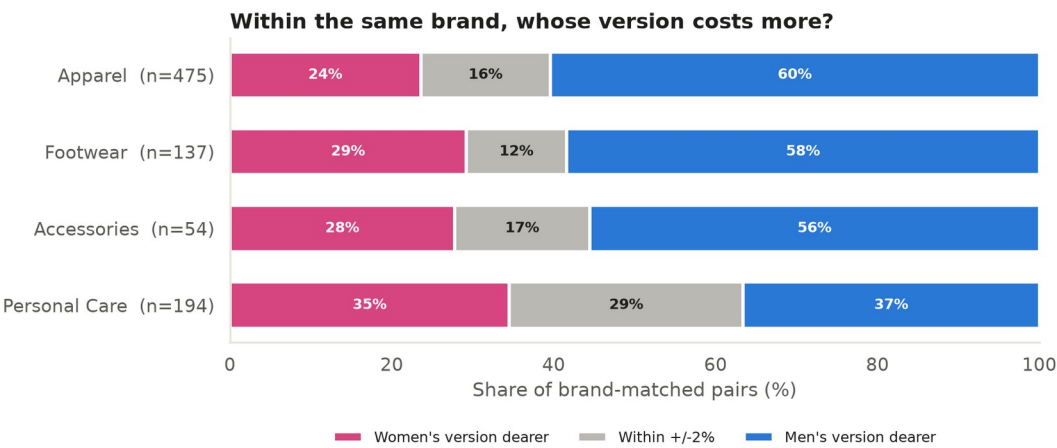


Figure 2.14: Share of pairs where each version is dearer, by segment

Day-12: Regression with Fixed Effects (Statsmodels)

Objective: estimate the gender premium while controlling for brand and category.

Activities performed

- Fitted OLS models of the form $\log(\text{MRP}) = \beta \cdot \text{Female} + \text{group effects}$, where $e^{\beta} - 1$ is the percentage premium.
- Used the *within* (demeaning) estimator for brand $\times$ category fixed effects and **clustered standard errors by brand**.
- Estimated separate models for each segment, for kids, and for price per 100 ml.
- Checked for a **composition effect**: kids' products look 18% dearer for girls overall, but the gap disappears within categories because girls' categories (kurta sets, clothing sets) are pricier in general.

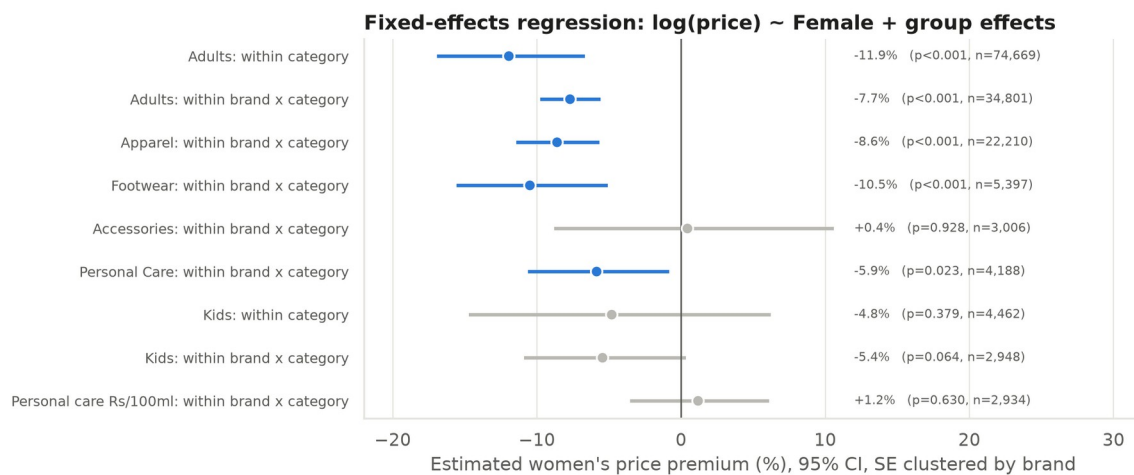


Figure 2.15: Fixed-effects estimates of the women's price premium

Result

Within the same brand and category, adult women's products are **7.7% cheaper** (95% CI -9.8% to -5.6%). Accessories show no gap (+0.4%, $p = 0.93$), and personal care per 100 ml shows no gap (+1.2%, $p = 0.63$). The kids' premium is not significant (-4.8%, $p = 0.38$).

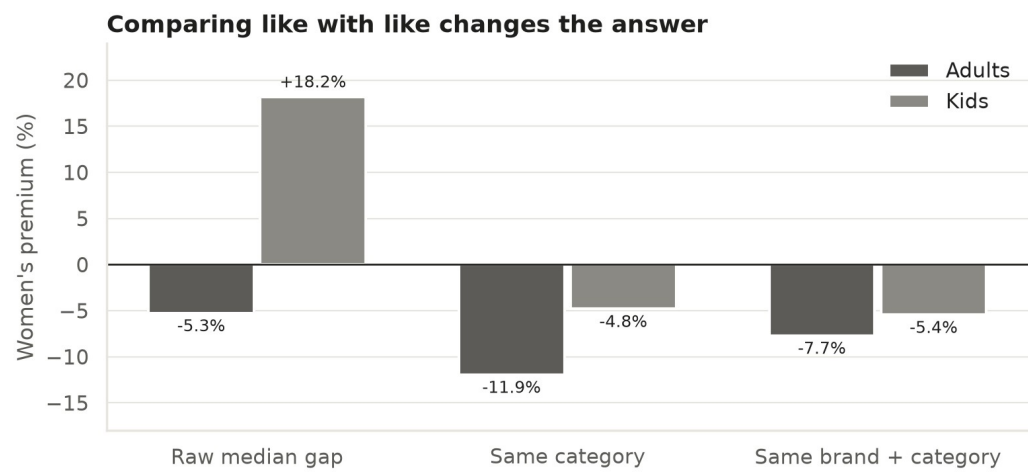


Figure 2.16: How the answer changes when like is compared with like

Day-13: Discount Analysis and Machine Learning (scikit-learn)

Objective: check whether discounts change the picture, and measure how much gender really explains price.

Activities performed

- Compared discounts. Women's items carry deeper discounts (**36.7% vs 32.9%**, $p < 0.001$), especially in apparel and accessories.
- Repeated the category tests on the **selling price**. The innerwear premium flips from +33% on MRP to -11% after discounts. BigBasket women's deodorants go from no gap to +17%, because they are discounted less. This is a Pink Tax at the point of sale.
- Trained a **random-forest** model (scikit-learn) to predict log price from category, segment, retailer, brand (target-encoded) and gender. It reached a test R^2 of 0.78.
- Used **permutation importance** to rank the features.

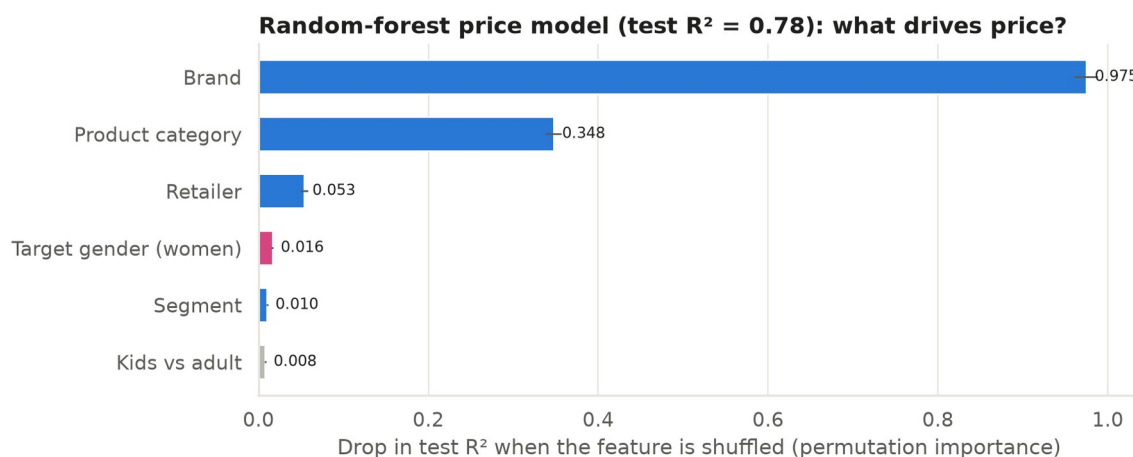


Figure 2.17: Permutation importance of each feature in the random-forest model

Result

Brand (importance 0.97) and category (0.35) explain most of the price. Target gender adds only 0.02. Price is driven by *what* the product is and *who makes it*, not by *who it is marketed to*.

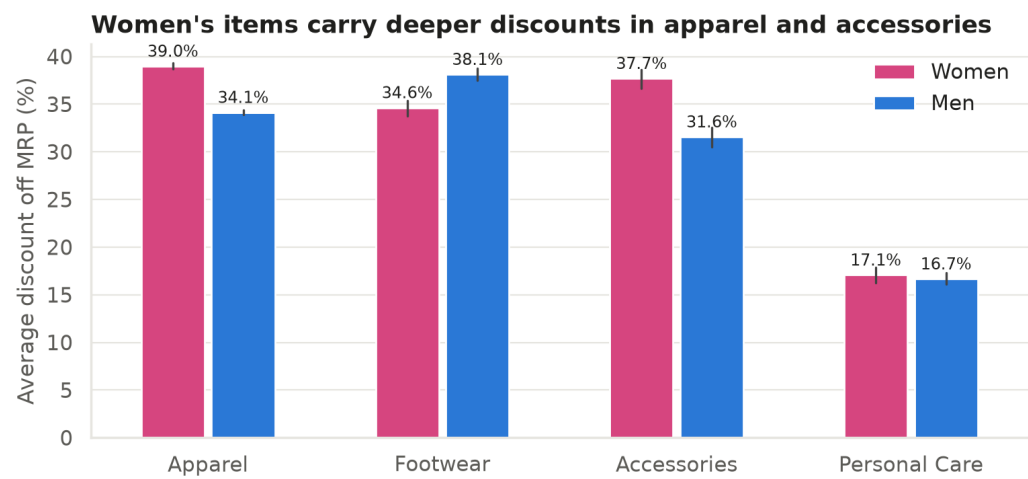


Figure 2.18: Average discount by segment and gender

Day-14: Excel Workbook – Formulas, Pivot Summaries and Charts

Objective: make the analysis available to Excel users without Python.

Activities performed

- Loaded all 79,390 products into an **Excel Table** (*tbl_Products*) so that it can be filtered, sorted and pivoted.
- Built a **Category_Summary** sheet with live **COUNTIFS** and **AVERAGEIFS** formulas, for example `=AVERAGEIFS(MRP, Source, A5, Category, D5, Gender, "Women")`, next to the medians, confidence intervals and p-values from Python.
- Built a pivot-style **segment × gender** summary, a **Matched_Pairs** sheet with data bars, and a **Stats_Tests** sheet.
- Applied **conditional formatting**: a blue-to-pink colour scale on the premium and coloured result labels.
- Added native Excel bar charts and checked that the formulas give the same numbers as Python.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	Category-level comparison													
2	<i>Columns E-J are live Excel formulas on the Data sheet; K-P come from the Python analysis.</i>													
3														
4	Source	Age group	Segment	Category	Products (Men)	Products (Women)	Avg MRP (Men)	Avg MRP (Women)	Avg-price premium %	Avg discount gap (pp)	Median MRP (Men)	Median MRP (Women)	Median premium %	95% CI low %
5	Myntra	Kids	Apparel	Night Suits	122	174	₹1,178	₹1,664	41.2%	12.4	₹899	₹1,599	77.9%	36.7%
6	Myntra	Adult	Apparel	Innerwear (Briefs)	408	488	₹671	₹768	14.5%	20.3	₹497	₹662	33.3%	24.1%
7	Amazon.in	Adult	Personal Care	Face Wash	95	31	₹508	₹524	3.2%	10.8	₹360	₹450	25.0%	0.0%
8	Myntra	Adult	Personal Care	Perfume	327	178	₹2,452	₹2,937	19.8%	-0.7	₹1,499	₹1,798	20.0%	-3.6%
9	Myntra	Kids	Footwear	Casual Shoes	121	119	₹1,435	₹1,595	11.2%	6.5	₹1,099	₹1,299	18.2%	-9.1%
10	Amazon.in	Adult	Personal Care	Body Wash	97	28	₹467	₹579	23.8%	-3.5	₹449	₹499	11.1%	-0.3%
11	Myntra	Kids	Apparel	Clothing Sets	216	291	₹1,395	₹1,642	17.7%	5.1	₹1,349	₹1,499	11.1%	4.9%
12	Myntra	Adult	Apparel	Sweaters	174	283	₹3,165	₹2,632	-16.8%	-0.9	₹2,299	₹2,499	8.7%	0.0%
13	BigBasket	Adult	Personal Care	Deodorant	329	171	₹380	₹620	63.3%	-1.0	₹279	₹299	7.2%	0.0%
14	Myntra	Adult	Accessories	Wallets	507	735	₹1,942	₹2,148	10.6%	4.7	₹1,699	₹1,799	5.9%	-3.4%
15	Myntra	Adult	Footwear	Sports Shoes	1988	882	₹2,862	₹3,383	18.2%	0.5	₹2,095	₹2,199	5.0%	-4.8%
16	Amazon.in	Adult	Personal Care	Shampoo	60	25	₹788	₹599	-24.0%	-0.4	₹522	₹539	3.3%	-29.9%
17	BigBasket	Adult	Personal Care	Perfume	140	96	₹1,738	₹1,700	-2.2%	1.2	₹1,297	₹1,300	0.2%	-29.7%
18	Myntra	Kids	Apparel	Kurta Sets	151	205	₹2,503	₹2,615	4.5%	-1.7	₹2,499	₹2,499	0.0%	-8.0%
19	Amazon.in	Adult	Personal Care	Deodorant	766	377	₹564	₹498	-11.7%	-3.1	₹349	₹349	0.0%	-5.4%
20	Myntra	Adult	Apparel	Kurtas	792	6023	₹2,144	₹2,185	1.9%	-0.2	₹1,999	₹1,996	-0.2%	-5.0%

Figure 2.19: Category_Summary sheet with formulas and conditional formatting

Learning outcome

Excel is still the most widely used analysis tool. Formulas that update when the data changes make a workbook reusable. Checking that the Excel formulas give the same numbers as Python is a quick way to validate both.

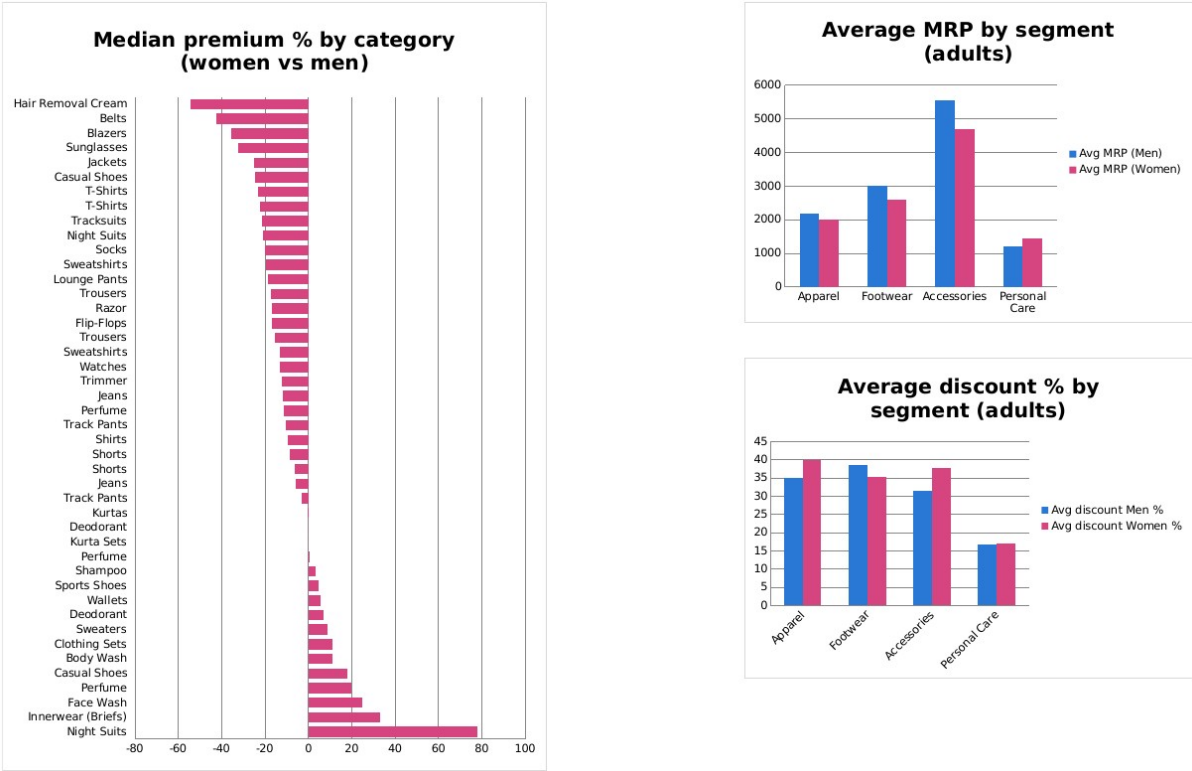


Figure 2.20: Native Excel charts

Day-15: Power BI Dashboard, GitHub Publishing and Report

Objective: build an interactive dashboard and publish the complete project.

Activities performed

- Exported Power BI-ready tables and designed a **star schema**: *fact_products* linked to *dim_category*, with *category_premium* and *brand_pairs* as result tables.
- Loaded them through **Power Query**, set the data types and created **DAX measures** such as *Median MRP (Women)*, *Pink Tax % = DIVIDE([Median MRP (Women)], [Median MRP (Men)]) - 1*, *Pairs Women Dearer %* and a conditional-colour measure.
- Built the dashboard: six KPI cards, a category gap chart, segment medians, a brand-pair outcome chart, a table of the Pink Tax pockets, and slicers for source, segment, age group and category. Applied a custom JSON theme.
- Wrote the README, pushed everything to GitHub and completed this report.

Learning outcome

A dashboard lets a non-technical user explore the result: selecting “Kids” or “Personal Care” in a slicer immediately updates every visual. Publishing the code and data on GitHub makes the whole study transparent and reproducible.

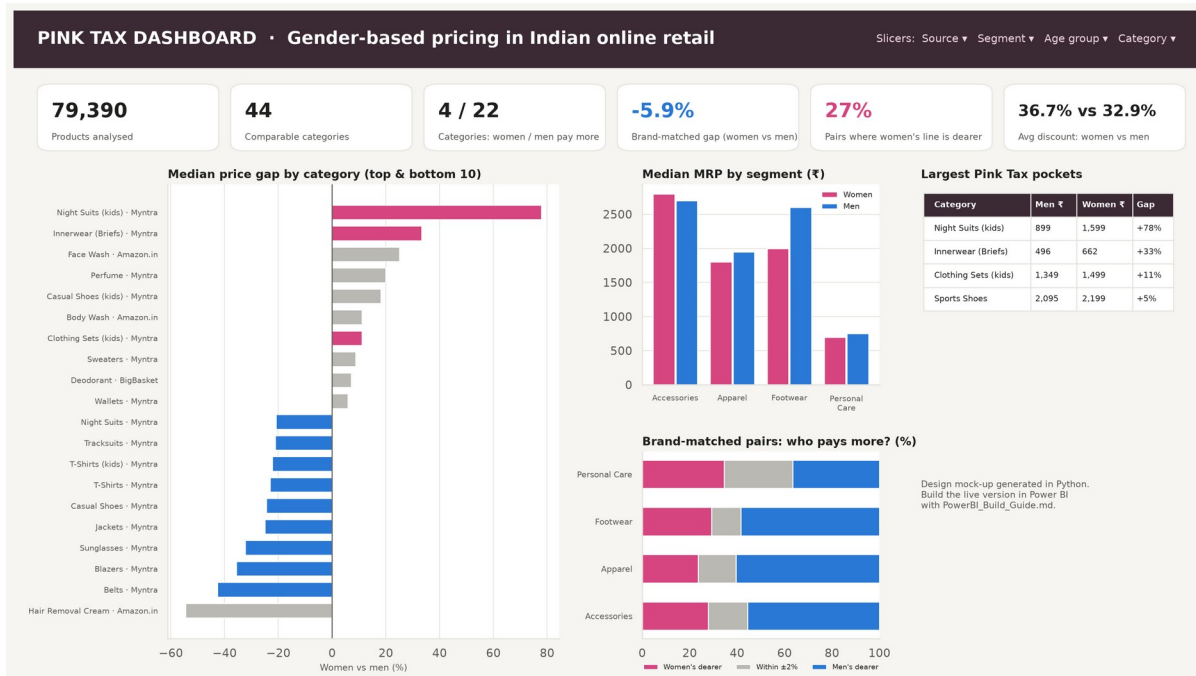


Figure 2.21: Pink Tax dashboard design (Power BI)

Chapter 3: Conclusion

3.1 Summary of Findings

This project set out to answer a simple question with real data: do products marketed to women in India cost more than comparable products marketed to men? After analysing 79,390 products from Myntra, BigBasket and Amazon.in, the answer is **“not in general, but yes in specific places.”**

1. **No blanket Pink Tax.** Women pay significantly more in 4 of 44 categories and men pay more in 22. Within the same brand and category, adult women's products are about 7.7% cheaper.
2. **The Pink Tax is concentrated** in girls' night suits (+78%), innerwear (+33%), kids' clothing sets (+11%) and sports shoes (+5%) at MRP, and in women's deodorants at the selling price (+17%).
3. **Like-for-like comparisons remove most gaps.** Personal care per 100 ml shows no significant difference, and the raw kids' premium disappears within categories.
4. **Brand and category drive price; gender barely does.** In the machine-learning model gender adds about 0.02 to R^2 , compared with 0.97 for brand.
5. **Discounts matter.** Women's items are discounted more deeply on average, so the MRP and the price actually paid can tell different stories.

These results agree with recent research that finds no systematic premium for identical products. The gaps reported in earlier studies mostly come from comparing differentiated products and different brand mixes. For consumers the practical advice is to compare **unit prices** and to look at the other gender's version of simple products such as razors, deodorants and basic clothing.

3.2 What I Learned from the Internship

- **Technical:** writing a complete, reproducible data pipeline in Python; advanced Pandas and regular expressions (including Hindi text); visualisation with Matplotlib, Seaborn and Plotly; Excel formulas and conditional formatting; Power Query, data modelling and DAX in Power BI; Git and GitHub.
- **Statistical:** choosing non-parametric tests for skewed data, bootstrap confidence intervals, multiple-testing correction, effect sizes, fixed-effects regression with clustered errors, composition effects (Simpson's paradox) and model interpretation with permutation importance.

- **Analytical thinking:** a headline number such as “women pay X% more” depends on *what is compared with what*. Checking assumptions and reading real examples matters as much as writing code.
- **Professional:** planning work in daily milestones, documenting decisions, and presenting findings honestly even when they differ from the popular expectation.

3.3 Limitations and Future Scope

- Gender and category tags are based on keywords, so a small share of products may be mislabelled.
- “Comparable” is approximated by same category and same brand. Differences in fabric, fit and design remain.
- The data are one-time snapshots (2022–2023) of listed prices and do not include sales volumes.
- **Future work:** track prices over time with web scraping, match exact twin products (same SKU except gender), add offline stores and services such as salons and dry-cleaning, and use text similarity models to find near-identical products automatically.

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